

Residual Momentum Across Markets

The hidden fragility of momentum in the United States

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ABSTRACT

Momentum means buying shares that have been rising and selling shares that have been falling. Residual momentum is a cleaned up version that first removes the part of each share's movement caused by the whole market, and ranks shares only on what they did on their own. It is widely believed to be the better, steadier version. This study asks a simple question: does it work equally well in every market? We test it in the United States and in Europe. It works in Europe and fails in the United States. We then spend the rest of the study asking why, testing one explanation at a time. It is not the sample period, because matching the years makes the United States worse. It is not the factor model, because removing five forces fails just as three did. It is not one sector, because removing all of technology changes nothing, and it is not the largest firms, because removing them changes nothing either. It is not luck, because across thousands of resamples Europe beats the United States almost every time. Every ordinary explanation fails, which pointed to something deeper than any single test could explain. Digging further, the United States signal turns out to have carried almost no genuine stock picking skill to begin with, only a bet on value stocks that happened to lose, while Europe's signal carried real, measurable skill. And in the calm years after the financial crisis, a single catastrophic month, August 2011, on its own accounts for most of the United States signal's loss, a month Europe's own signal came through largely unaffected. With only two markets and data that ends in 2018, we do not claim to know why the United States lost its skill in this window, or exactly what happened that month, and we are honest about leaving those two questions open. The way to settle them is to widen the

study, testing residual momentum across more markets and on current data, and then comparing the markets where it works against those where it fails. The broad lesson is already clear: removing the market is not a free improvement everywhere, so the cleaned up version cannot be assumed to be the safer, steadier choice in every market.

1

The Research Question

Does the cleaned up version of momentum work in every market, or only in some?

Momentum is one of the oldest and most reliable ideas in investing (Jegadeesh and Titman, 1993). Shares that have gone up over the past year tend to keep going up for a while, and shares that have gone down tend to keep going down. So an investor who buys the recent winners and sells the recent losers tends, on average, to make money. This pattern has shown up across many countries and many decades (Rouwenhorst, 1998; Asness, Moskowitz, and Pedersen, 2013).

A few years ago researchers suggested an improvement. They pointed out that part of a share's recent rise was never really about the company. When the whole market goes up, it carries almost every share with it, the strong and the weak alike. So they built a cleaned up version, called residual momentum, that first removes the market's push and then ranks each company only on what it did on its own. The promise was a cleaner, calmer, more dependable signal.

This study asks one plain question. Does that cleaned up version actually work better everywhere, or only in the markets where it was first tested? We answer it by building the same strategies in two of the world's largest markets, the United States and Europe, and comparing them carefully. To see why the cleaned up version is expected to be better, and therefore why it is surprising when it fails, we start with a short background.

2

A Short Background

What momentum is, and why the cleaned up version is trusted.

Ordinary momentum simply buys recent winners and sells recent losers (Jegadeesh and Titman, 1993). Its profits are not fully explained by the usual sources of risk (Fama and French, 1993; Carhart, 1997), which is why it is treated as a force of its own. It also has a well known weakness. Every so often it crashes hard. This usually happens when a falling market suddenly turns

upward and the beaten down shares that the strategy is betting against come roaring back (Daniel and Moskowitz, 2016).

Residual momentum was the proposed fix (Grundy and Martin, 2001; Blitz, Huij, and Martens, 2011). The idea is to remove the part of each company's return that came from the market and a few other broad forces, and to keep only what is left, the part that belongs to the company itself. A simple way to picture it is an exam grade adjusted for how hard the exam was. Seventy out of a hundred on a very hard paper is more impressive than seventy on an easy one. Residual momentum makes that kind of fairness adjustment for every company, every month. Those earlier studies, carried out mainly in the United States, found it worked well. What nobody had checked carefully is whether it works equally well in different markets, and that gap is what this study fills.

3

The Data

Two markets, and a deliberate choice to keep the companies that failed.

We use monthly data for the companies in two well known indices. The S&P 500 stands in for the United States and the Stoxx 600 stands in for Europe.

One choice matters more than any other. We did not only use the companies that are in each index today. We used every company that was ever in each index during the study, including the ones that later shrank, merged, or went bust. If you leave out the failures, the past looks far safer and more profitable than it really was, because the disasters quietly vanish from the record. Keeping them in keeps the study honest.

Table 1. The two markets we study

Market	Years of data	Companies
United States, S&P 500	1990 to 2018	1,355
Europe, Stoxx 600	2002 to 2018	1,255

Why this matters. *The United States data start earlier, simply because the European records begin later. That difference in length looks harmless here, but it becomes the centre of the first test later on.*

4

How We Built and Measured the Strategies

Three strategies, written out plainly, and the four numbers we judge them by.

We build three strategies in each market. Each one is rebuilt every month, and each one buys the strongest tenth of companies and sells the weakest tenth.

The first is **plain momentum**. Each month we score every company by how much it rose or fell over the past year, but we skip the most recent month, because very recent moves often reverse and would mislead us. In symbols, the score for company i in month t is:

$$M(i,t) = (1 + r(i,t-12)) \times (1 + r(i,t-11)) \times \dots \times (1 + r(i,t-2)) - 1$$

Reading that line in plain words: $M(i,t)$ is the momentum score for company i in month t . The letter r is a monthly return, so $r(i,t-12)$ is the company's return twelve months ago. The line multiplies the returns from twelve months ago up to two months ago, which gives the company's total gain over that stretch. We buy the highest scores and sell the lowest.

The second is **residual momentum**. It does the same ranking, but first it removes the market's push. This happens in two steps. In step one, for each company we run a simple fit over the last thirty six months that splits the company's return into the part explained by three broad forces and the part left over:

$$r(i,s) - r_f(s) = a + b_1 \times \text{Market} + b_2 \times \text{Size} + b_3 \times \text{Value} + e(i,s)$$

Reading that line: the left side is the company's return above the safe, risk free rate. Market, Size and Value are the three broad forces, explained just below (in standard notation, Market is $MKT_t - R_{f,t}$, Size is SMB_t , and Value is HML_t). The numbers b_1 , b_2 and b_3 measure how strongly each force moved this particular company. And e , the final term, is the residual, the part of the return that none of the three forces can explain. That leftover is the company's own performance, which is exactly what we want to rank on.

The three forces we remove, known together as the Fama and French three factor model, are these. **Market** is the whole market rising or falling, which lifts

or drops almost everything at once. **Size** is the tendency for small companies to behave differently from large ones. **Value** is the tendency for cheap companies, priced low relative to their books, to behave differently from expensive ones. Removing all three leaves only what the company itself did.

In step two, we take that leftover for the last twelve months, skipping the most recent month, and turn it into a score:

$$RM(i,t) = (\text{average of the leftover } e \text{ over those months}) / (\text{how much the leftover wobbled})$$

Here $RM(i,t)$ is the residual momentum score. The top of the fraction is the company's average leftover return. The bottom is how much that leftover bounced around from month to month. Dividing one by the other rewards steady, dependable strength and punishes lucky, jumpy gains.

The third is **volatility targeted momentum**. It keeps the plain momentum picks but changes how much money goes in, depending on how calm or wild markets have been:

$$w(t) = (10 \text{ percent annualised, the target volatility}) / (\text{how wild the market has been recently, capped so } w \text{ never exceeds } 1.5)$$

In that line, $w(t)$ is how large the position is in month t . The target risk level is a fixed amount of risk we choose to aim for. When markets are calm, the bottom of the fraction is small, so w is large and we invest more. When markets are stormy, the bottom is large, so w is small and we pull back. The aim is to be holding less right when a crash is most likely.

We judge all three on the same four numbers;

The annualised return is the average gain per year.

The annualised volatility is how much the monthly returns bounced around, written as a yearly figure; a smaller number means a steadier ride.

The Sharpe ratio is the return divided by the volatility, so it is the reward earned for each unit of risk, and higher is better.

The maximum drawdown is the deepest fall from a previous high, which tells you the worst stretch an investor would have had to live through.

A few practical details make the study reproducible. Returns are computed from the supplied monthly price series; the treatment of dividends and corporate actions follows the data provider's own series (see Appendix B). The companies come from a single **survivorship free** dataset of index members that keeps every firm that was ever in each index, including those that later left, merged, or failed. Each month the strongest and weakest tenths are held in **equal weight**, so every chosen company counts the same, and the portfolios are rebuilt fresh each month rather than overlapped. When a company has no return in a given month, for example before it joined the index or after it left, it is left out of that month's ranking rather than filled in with a guess.

5

The First Look

Each market measured over its own full history.

We start by measuring each market over all the years it has. Plain momentum makes money in both. It returns 4.4 percent a year in the United States and 14.9 percent in Europe. So far the two markets agree, and Europe is simply the stronger of the two.

Table 2. How each strategy did over its own full history

Strategy	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown
United States, plain	4.4%	29.4%	0.31	81.9%
United States, residual	1.6%	18.3%	0.20	74.0%
United States, volatility targeted	8.7%	14.4%	0.65	35.3%
Europe, plain	14.9%	23.4%	0.74	74.1%
Europe, residual	5.2%	9.1%	0.60	26.2%
Europe, volatility targeted	16.6%	13.7%	1.20	27.3%

Reading the numbers. Annualised return is the average yearly gain. Annualised volatility is how much the returns bounced around, as a yearly figure; lower is steadier. The Sharpe ratio is return divided by volatility, so it is reward for each unit of risk. Maximum drawdown is the deepest fall from a previous high. Bigger return and Sharpe are better, smaller volatility and drawdown are better.

Now look at the cleaned up version, residual momentum, in each market. In Europe it does exactly what it promises. It keeps a healthy return of 5.2 percent

a year, but much steadier, with a volatility of only 9.1 percent and the smallest worst fall of any European strategy, 26 percent. In the United States the same cleaning leaves a return of just 1.6 percent a year and a Sharpe ratio of 0.20, the weakest result on the whole board apart from its own plain version. Europe behaved as the textbook says. The United States did not.

One detail in this same table is worth pausing on. In the United States, the two stock-selection strategies, plain and cleaned up, were the weak performers, yet the third strategy, volatility targeted momentum, was the strongest of all, with a Sharpe ratio of 0.65. That strategy does not pick stocks any more cleverly; it simply adjusts how much is invested as markets calm down or heat up. It is a striking early hint that in the United States, improving momentum may depend more on managing risk through time than on filtering stocks more finely.

6

The Surprise

Cleaning the signal helped Europe but hurt the United States.

This is the opposite of what was supposed to happen. The whole point of removing the market is to make the signal cleaner and therefore better. In Europe it did. It raised the reward for risk and roughly halved the worst fall. In the United States it did the reverse. It stripped the return down to almost nothing and left a strategy that barely pays you for the risk you are carrying.

The difference is too large to ignore. The same method, using the same forces, improved momentum in one market and damaged it in the other. Something has to explain that. The rest of this study tests three explanations, one at a time, in the order a careful person would try them. The first and most obvious is that the United States simply has more years of data than Europe. We test that next.

7

First Test: Is It Just the Extra Years?

The United States history is longer. Could that alone be the cause?

The most obvious explanation is the difference in the length of the two histories. The United States data run from 1990. The European data effectively begin in 2006, because residual momentum needs three years of history before it can produce its first score. So the United States carries an extra stretch of the 1990s, which was a very good decade for momentum, plus more booms and busts along the way. Maybe that extra, kinder history is doing all the work.

There is a clean way to check. We cut both markets down to the exact months they share, from 2006 to 2018, and measure again. This is what we mean by the matched window: the same set of calendar months for both markets, so the comparison is finally fair. Because Europe already lives almost entirely inside those years, the cutting falls almost entirely on the United States, which loses its earliest and best years. We expected this to help the United States. If its weakness was only caused by having more, kinder years, then taking those years away should pull its result back up towards Europe.

Table 3. The same strategies, now over the shared years, 2006 to 2018

Strategy	Annualised return	Annualised volatility	Sharpe ratio	Maximum drawdown
United States, plain	-3.8%	25.6%	0.00	81.5%
United States, residual	-7.1%	20.4%	-0.23	74.0%
United States, volatility targeted	4.4%	13.1%	0.40	35.3%
Europe, plain	12.8%	24.9%	0.64	74.1%
Europe, residual	5.2%	9.1%	0.60	26.2%
Europe, volatility targeted	13.2%	13.5%	0.99	27.3%

Key observation. *The United States numbers fall sharply, while the European numbers barely move. European residual momentum is exactly the same as before, because its full history already was these years.*

The result is the opposite of what we expected. Cutting the United States back to the same years did not help it. It made it worse. Plain momentum in the United States dropped to a Sharpe ratio of zero, and residual momentum, which was merely weak before, turned firmly negative at minus 0.23. Europe barely

changed, because for Europe almost nothing was removed. So having more years of data was not the cause. The clearest way to see this is to put each market's full history next to its matched window, side by side.

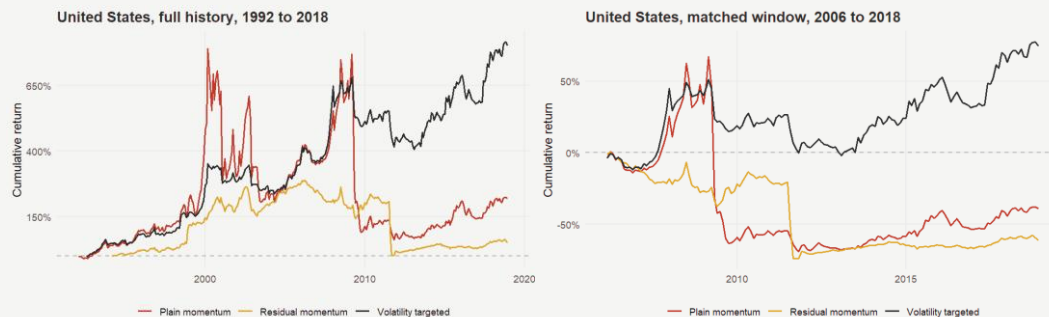


Figure 1. Cumulative return chart. United States: full history on the left, matched window on the right. Each line shows what one unit of money grows into over time. The dashed line in the middle is the starting point: above it means a profit, below it means a loss.

These two pictures are the same three United States strategies, and they look almost nothing alike. On the left, over the full history, all three lines end above the starting line, so all three made money in the end, although the residual line is much weaker than the others and fell sharply more than once along the way. On the right, over the matched years only, the picture changes completely. The plain and residual lines drop and finish well below the starting line, so they lost money, and only the volatility targeted line stays up. Nothing about the strategies changed between the two pictures. Only the years did. The healthier picture on the left was being held up by the 1990s and early 2000s, the very years Europe never had. Take those years away, and what is left is the picture on the right, where United States momentum loses money.

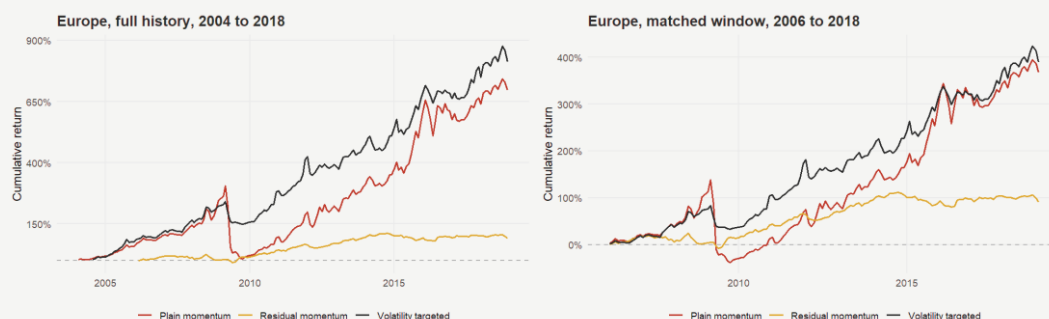


Figure 2. Cumulative return chart. Europe: full history on the left, matched window on the right. The same growth of one unit of money, for Europe, whose data already begin close to the start of the shared years.

For Europe the two pictures tell almost the same story, which is exactly what we would expect. Europe's data already begin around 2006, so cutting to the shared years removes very little. On the left, over the full history, the plain and volatility targeted lines climb strongly and the residual line rises gently but steadily into profit. On the right, over the matched years, the shape is almost identical, just a little shorter, and the residual line is essentially unchanged. Where the United States was transformed by matching the years, Europe was barely touched.

Putting the two markets next to each other makes the point plain. If the difference between them were only about which years were included, then matching the years would have changed both markets. Instead it rewrote the United States and left Europe almost exactly as it was. We can also see the same effect across all six strategies at once in two bar charts, where each strategy gets two bars: one for its full history and one for the matched years.

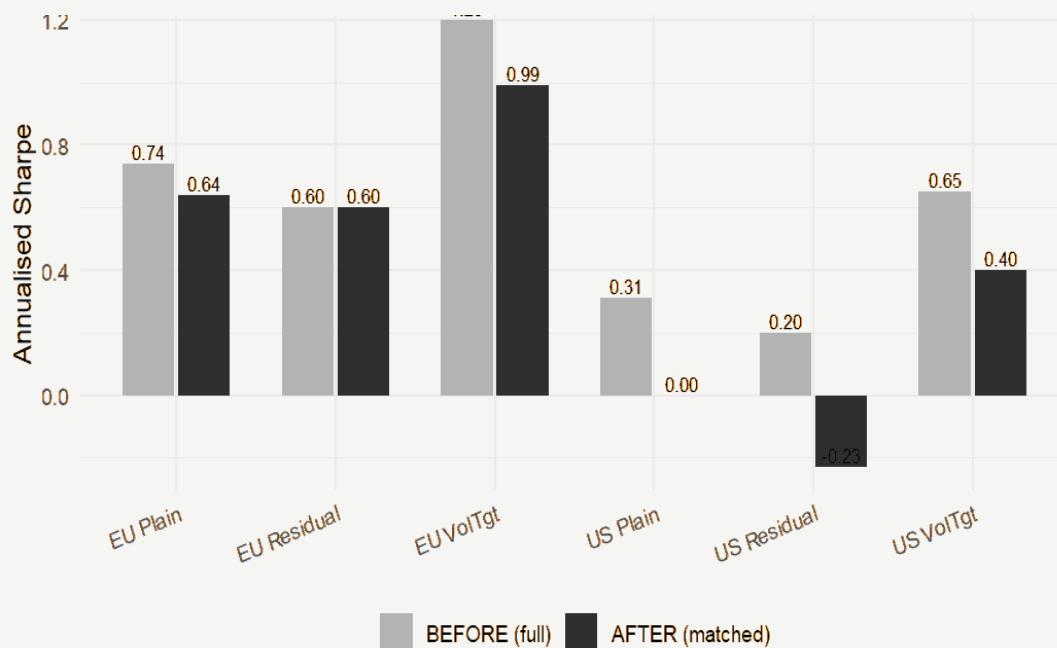


Figure 3. Grouped bar chart of the Sharpe ratio, full history versus matched years. Each strategy has two bars. The lighter bar is the full history; the darker bar is the matched years. You read it by comparing the two bars in each pair.

In this chart, the European pairs are almost equal in height, so matching the years changed Europe very little. The United States pairs drop sharply, and the United States residual bar falls below the zero line, which means it lost money

once the kinder early years were removed. The chart says in one picture what the table says in numbers: matching the years harmed only the United States.

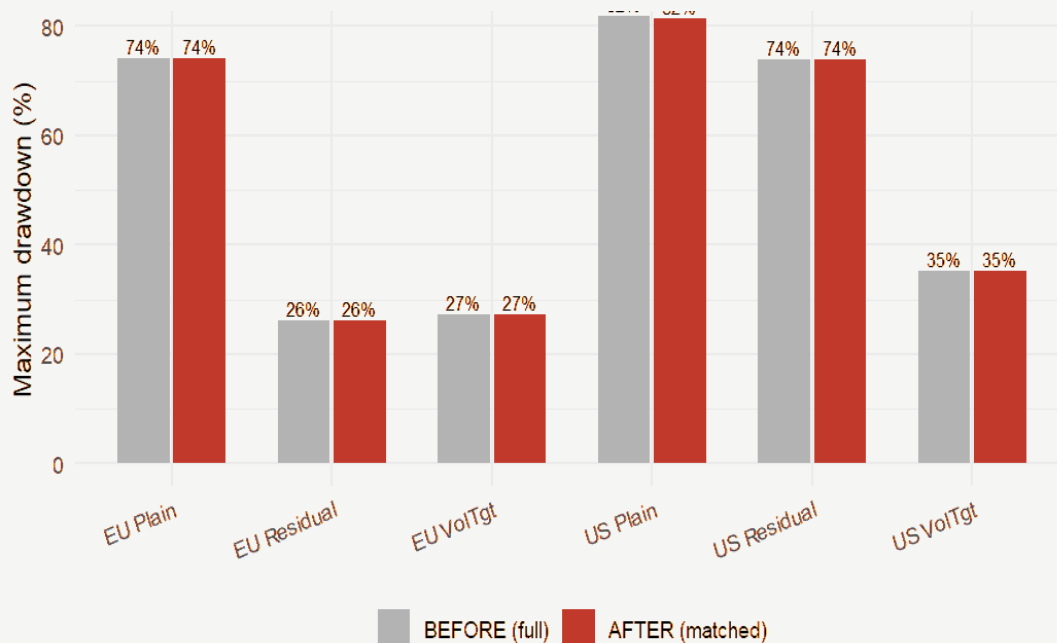


Figure 4. Grouped bar chart of the maximum drawdown, full history versus matched years. Again two bars per strategy, the full history and the matched years, with shorter bars meaning a gentler worst fall.

Here almost nothing moves. The worst falls are about the same before and after matching, in both markets. The reason is simple: every strategy's deepest fall happened during the 2008 crisis, and 2008 is inside the shared years. Cutting the early years away removed good years, not the crash. So the crashes were never going to change, and they do not. The conclusion of this first test is clear. The United States weakness is not caused by having more years of data. If the period is not the reason, the next thing to check is the market itself.

8

Second Test: Is It the Market Itself?

Once the market is removed, what is left, and can it be trusted?

Ruling out the years sharpens the question. If the difference is not about the period, perhaps it is about the market.

By the market we mean the three broad forces that move shares regardless of how any single business is really doing: the **market** itself, the tide that lifts or lowers almost every company at once; **size**, the tendency for small companies to behave differently from large ones; and **value**, the tendency for cheap companies to behave differently from expensive ones. These are the three forces the cleaned up version removes. The cleaned up version gives us a direct way to see how much those forces mattered, because the only difference between plain and residual momentum is whether they are left in or taken out. Comparing the two therefore measures exactly what they were contributing in each place.

How much was the market adding?

Table 4. Yearly return with the market left in, and with it removed, over the shared years

	United States	Europe
Plain momentum, market left in	-3.8%	12.8%
Residual momentum, market removed	-7.1%	5.2%
The difference, what the market was adding	3.4%	7.6%

What changes after removing the market? *The difference between the two rows is roughly how much the market was contributing. In Europe a healthy return remains once the market is removed; in the United States the return falls further into loss.*

Two things stand out. In Europe, removing the market still leaves a healthy 5.2 percent a year, so the European companies carry the signal on their own. In the United States, removing the market lowers the return from minus 3.8 to minus 7.1 percent, so the companies on their own are a drag. The difference row adds a detail that runs against the obvious story. The market actually gave more to Europe, 7.6 percent, than to the United States, 3.4 percent. So the United States problem is not a generous market hiding weak companies. Even with the market's smaller help, the United States signal lost money, and without that help it lost more.

Could the signal be trusted?

Return alone does not tell the whole story, though. A strategy can make money while still being unreliable, and it can lose money while still being steady. So the next table asks a different question: not how much the United States signal earned, but whether it was steady enough to trust from one month to the next.

Table 5. How dependable the cleaned up signal was, over the shared years

	United States	Europe
Share of months that ended positive	51.3%	61.7%
Typical swing from one month to the next	5.9%	2.6%
Worst single month	-48.0%	-6.8%

What these numbers tell us. *A signal that is positive only about half the time is close to a coin toss. A larger month to month swing, and a deeper worst month, both mean a less dependable signal.*

In the United States the cleaned up signal was positive in only 51.3 percent of months. That is barely better than tossing a coin. It also swung more than twice as much from month to month as the European signal, and in its worst single month it fell by 48 percent, almost half its value, against a worst European month of under 7 percent. A signal that has supposedly been cleaned of market noise, yet is still close to a coin toss and can lose half its value in one month, is hard to treat as real information. Both the size and the steadiness of the United States signal point the same way, and away from the calendar.

Putting it to a formal test

So far these figures describe what happened. A fair question is whether they could simply be the result of chance. To check, we use a **significance test**. The idea is simple. We begin by assuming the boring position, that nothing special is happening, which is called the **null hypothesis**. We then ask how likely our result would be if that boring position were true, a likelihood called the **p value**. A small p value, below 0.05, means the result would be very unlikely by chance, so we treat it as real. A large p value means the result could easily be luck, so we cannot claim anything. We ran two tests on the shared years.

The first test asks whether the share of positive months is really different from a coin toss. The null hypothesis is that the signal is positive half the time, purely by chance. For the United States the share was 51.3 percent, with a p value of

0.81. That is far above 0.05, so we cannot reject the coin toss: the United States signal is, in statistical terms, **a coin toss**. For Europe the share was 61.7 percent, with a p value of 0.005, well below 0.05, so Europe is positive more often than chance would allow.

Stated in the usual form: the null hypothesis, written **H0**, is that the signal is positive half of the time, purely by chance; the alternative, written **H1**, is that it is positive more often than half of the time. This is a test of a proportion against 0.5. For the United States the p value of 0.81 keeps **H0**; for Europe the p value of 0.005 rejects **H0** in favour of **H1**.

The second test asks whether the average monthly return is really different from zero. The null hypothesis is that the true average is zero, meaning no real profit. For the United States the average was minus 0.39 percent a month, which could not be told apart from zero, with a p value of 0.41. For Europe the average was plus 0.45 percent a month, a genuinely positive figure, with a p value of 0.03.

Stated in the usual form: the null hypothesis **H0** is that the true average monthly return is zero; the alternative **H1** is that it is not zero. This is a one sample t test on the monthly returns. For the United States the statistic is t equals minus 0.82 with a p value of 0.41, which keeps **H0**; for Europe it is t equals 2.14 with a p value of 0.03, which rejects **H0** in favour of **H1**. Because monthly momentum returns can be autocorrelated and heteroskedastic, both t-tests were re-checked using Newey-West (HAC) standard errors (Appendix B); the conclusion for each market was unchanged.

Together these two tests confirm the picture in plain terms. Europe's signal is real: it is positive more often than chance, and it earns a return that is genuinely above zero. The United States signal behaves like pure chance: it is as likely to be positive as a coin toss, and its average return cannot be told apart from zero. The plain description and the formal test now agree. One market has a real signal, and the other has only noise.

A stronger check: reshuffling the years

Two tests pointing the same way is reassuring, and it leads naturally to one more question. What if the whole result rests on a single run of history, and

these particular years simply happened to be unkind to the United States? There is a way to check, called the bootstrap. It takes the real monthly results and reshuffles them into thousands of slightly different versions of the same period, then works out the outcome in every one. If the United States comes out weak in almost all of those thousands of versions, its weakness is not the bad luck of one stretch of time. It is built into the data.

Table 6. What thousands of reshuffles say about each market, over the shared years

	United States	Europe
Sharpe ratio, most likely value	-0.19	0.65
Sharpe ratio, 95 percent range	-0.62 to 0.52	0.06 to 1.28
Yearly return, 95 percent range	-19.1% to 5.8%	0.6% to 11.2%

What the ranges tell us. Look at whether zero sits inside each range. If zero is inside, then nothing is still a believable answer, so the signal is not proven. If zero is outside, the signal is genuinely there. Europe's range lies entirely above zero; the United States range includes zero.

There is a simple and standard way to read a range like this, and it is worth stating plainly because it is easy to get the wrong way round. A range is the set of values the true answer could reasonably be. The question is simple: does the range include zero, or does it stay entirely on one side of it? If the range still includes zero, then zero, meaning no real signal at all, is one of the answers we cannot rule out, so we cannot say the strategy is truly different from nothing. If the whole range sits on one side of zero, then zero has been ruled out, and the result is real. By that rule, Europe's range of 0.06 to 1.28 sits entirely above zero, so Europe's signal is genuinely real. The United States range of minus 0.62 to plus 0.52 still includes zero, so the United States signal cannot be separated from nothing. The gap between the two is real as well: across five thousand reshuffles, Europe came out ahead of the United States in 97.8 percent of them, and the United States signal fell below zero in 71.7 percent. The weakness is not the bad luck of one path through time.

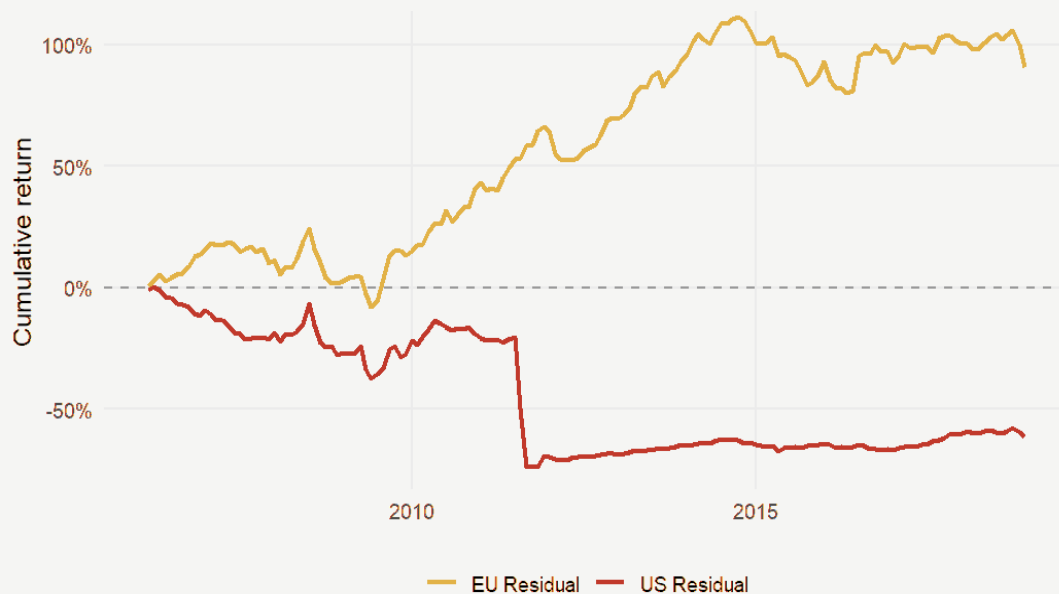


Figure 5. Cumulative return chart. The cleaned up signal in both markets, over the shared years. Growth of one unit of money for the residual strategy alone, in each market, on the same set of months.

This last picture shows the two cleaned up signals side by side over the same years. The European line climbs steadily into profit, while the United States line sinks and stays there. The same method, over the same months, ends in opposite places. That points to a real difference between the two markets, not to anything about the time period. So the second test gives a firm answer: the difference is structural, and it does not come from the period we happened to study. One question remains, and a careful reader would ask it. What if we removed the market in too simple a way?

9

Third Test: Is It the Way We Removed the Market?

We removed three forces. What if we remove five?

So far, removing the market has meant removing three broad forces: the market, company size, and value. This three force recipe is the standard one, known as the Fama and French three factor model. But it is not the only choice. A fuller recipe, the five factor model, removes those same three and adds two more. A fair challenge to our result is this: maybe three forces was too rough a

clean, and some leftover junk in the United States is what behaved badly. If we clean more thoroughly with five forces, does the United States signal recover?

The two extra forces in the five factor recipe are these;

Profitability is the tendency for companies with strong, steady profits to behave differently from companies with weak profits.

Investment is the tendency for companies that grow cautiously to behave differently from those that expand aggressively.

Removing these two as well leaves a cleaner leftover than before. We expected that the stronger clean might rescue the United States, and we tested it on the matched years, the ones that matter for our conclusion.

Table 7. Removing three forces versus five forces, over the shared years

Cleaned up signal	Annualised return	Sharpe ratio	Worst fall
United States, three forces removed	-7.1%	-0.23	74.0%
United States, five forces removed	-2.3%	-0.03	54.0%
Europe, three forces removed	5.2%	0.60	26.2%
Europe, five forces removed	3.7%	0.44	30.0%

What changes after a stronger clean? Removing two extra forces lifts the United States, but not above zero. Europe slips a little, because it was already clean and the extra removal trims some real signal away.

The stronger clean does help the United States. Its return rises from minus 7.1 to minus 2.3 percent a year, its worst fall shrinks from 74 to 54 percent, and its Sharpe ratio improves from minus 0.23 to minus 0.03. So part of the United States weakness really was the rougher three force clean leaving some junk behind. But notice where it ends. Minus 0.03 is still not a winning signal. After a fuller, fairer clean, the United States residual still earns essentially nothing. Europe moves the other way, from 0.60 down to 0.44, and it is worth being precise about what that drop means. When removing the two extra forces lowers Europe's result, it shows that part of Europe's edge was being paid for exposure to profitability and investment, which a five factor view treats as known, rewarded risk rather than pure stock selection. That does not weaken the comparison. If anything it strengthens it, because even after those forces are

removed from both markets, Europe stays clearly positive at 0.44 while the United States stays below zero at minus 0.03.

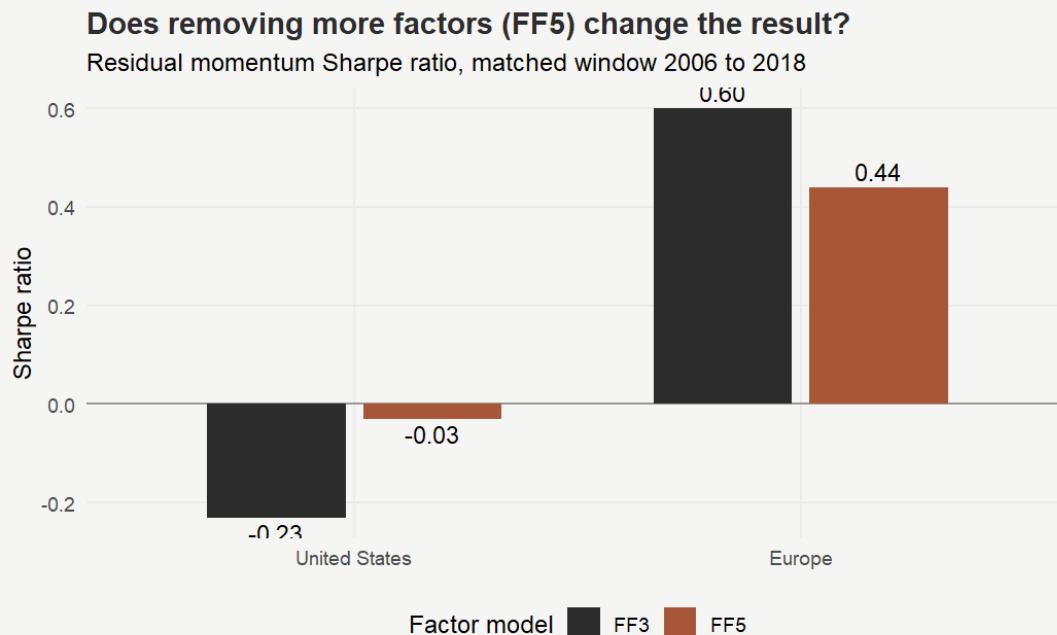


Figure 6. Bar chart of the Sharpe ratio under each recipe, over the shared years. Two bars for each market: the darker bar removes three forces, the lighter bar removes five. Above the zero line is a winning signal, below it is a losing one.

The picture makes the answer plain. Both United States bars sit at or below the zero line, while both European bars sit well above it. Removing more forces lifts the United States bar, but not over the line. This is the outcome that makes the finding solid. We removed the market in two different ways, the standard three forces and the fuller five, and the United States signal failed both times, while the European signal stayed positive both times. So the United States weakness is not an accident of one particular recipe. It survives a stronger clean. The difference is not the time period, and it is not the method. What we have not yet asked is when, and out of what raw material, the United States failure is made — the question the rest of this study turns to next.

10

What It All Means

Three tests, one consistent answer.

The three tests point the same way. Matching the years made the United States worse, not better, so the difference is not the time period. Removing the market left a losing, unreliable signal in the United States and a healthy one in Europe. And removing the market more thoroughly did not rescue the United States, so the difference is not the recipe either. When the same method, over the same months, cleaned the same way, leaves a clean profit in one market and a coin toss in the other, what remains is a structural difference between the two markets that none of these explanations can account for. So we asked two different questions instead: what was the United States signal made of, and when exactly does it break?

Why might the two markets differ?

Several reasons could explain why the two markets differ, and our data cannot separate them all at once, but they are worth naming because the rest of this section tests them one by one. The first and most obvious suspect is concentration. The United States market is dominated by a small number of very large companies, such as Apple, Microsoft, Nvidia, Amazon, and Meta, and it is natural to wonder whether a few giants moving together crowd out the company by company variety a residual signal needs. This is the first thing a careful reader would suspect, and it is the one we can test directly by removing them. A second possibility is that the United States momentum trade has grown crowded, with many investors chasing the same shares, which tends to make any strategy fragile. A third is that the market's push is itself part of what makes momentum work in the United States, so that removing it strips out part of the reward rather than cleaning the signal. We take the testable suspect first.

Testing the Leading Suspect: Removing Technology

Because concentration was the leading suspect, and the one our data could test directly, we put it to the test. If a handful of giant technology companies were what breaks the United States signal, then taking Technology out of the market

entirely should let the signal recover. So we removed every technology company, 197 of them in the United States and 57 in Europe, rebuilt the cleaned up signal on what remained, and measured again.

Table 8. The cleaned up signal with the technology sector removed, over the shared years

Cleaned up signal	Yearly return	Sharpe ratio	Worst fall
United States, all stocks	-7.1%	-0.23	74.0%
United States, technology removed	-8.0%	-0.19	80.5%
Europe, all stocks	5.2%	0.60	26.2%
Europe, technology removed	4.7%	0.55	25.3%

What changes after removing technology? *Almost nothing. The United States signal stays negative and Europe stays positive, so the technology sector is not what separates the two markets.*

Removing Technology did not rescue the United States. Its cleaned up signal stayed firmly negative, a Sharpe ratio of minus 0.19 against minus 0.23 before, and its return did not improve. Europe barely moved. So the leading suspect turns out not to be guilty, at least not on its own. The United States weakness is not the work of one fashionable sector, it is spread across the whole market. That is a stronger result than if Technology had been the culprit, because it means the failure is broad rather than a quirk that would vanish if a few names were dropped. A finer version of this test, removing only the very largest firms month by month, would need company size data we do not have here, and is left for future work.

Testing it further: removing the largest firms

Removing a whole sector was one way to test concentration. There is a sharper way. The real worry was never technology in particular, it was that a few enormous companies, whatever sector they sit in, dominate the United States market and drown out everything else. So we tested that directly. Each month we ranked every company by its size, using its weight in the index, removed the largest 10 percent by that measure, and rebuilt the cleaned up signal on the rest. If the giants were the problem, taking them out should let the United States signal recover.

Table 9. The cleaned up signal after removing the largest firms, over the shared years

Cleaned up signal	Yearly return	Sharpe ratio	Worst fall
United States, all stocks	-7.1%	-0.23	74.0%
United States, largest firms removed	-8.2%	-0.25	78.1%
Europe, all stocks	5.2%	0.60	26.2%
Europe, largest firms removed	5.1%	0.58	30.2%

What changes after removing the largest firms? *Again almost nothing. The United States signal stays negative and Europe stays positive, so the biggest companies are not the cause either.*

The result is the same as before. Removing the largest United States firms did not lift the signal; it stayed negative, at a Sharpe ratio of minus 0.25, and removing a smaller top slice made no difference either. Europe was barely touched. So concentration in the giant firms is not what breaks the United States, any more than the technology sector was. We have now ruled out the two explanations a reader would reach for first, one sector and the biggest names, and neither is the answer. The gap is real, it is robust, and at this point it is unexplained. So we stopped testing suspects and changed the kind of question we were asking.

Testing where the market's contribution actually came from

The first new question was not about the cleaned-up strategy at all. It was about the raw material. Table 4 already showed a difference between two numbers, but a difference between two numbers is only an inference about where the money came from. A more direct check takes each market's uncleaned momentum portfolio on its own and asks a sharper question of it: once the same three forces are accounted for, is there a genuine return left over that they cannot explain, called alpha, or was the return simply a side effect of leaning on one of the three forces, called a factor bet? Alpha is what a fund manager would want to be paid for: skill. A factor bet is closer to a coin that happened to land the right way up. We answered this by taking each market's monthly plain momentum return and running the same three factor comparison used throughout this section directly on it, over the shared years.

Table 10. What plain momentum's own return was made of, over the shared years

	United States	Europe
Alpha — the return left over after the three forces, annualised	0.0% a year, not distinguishable from zero	15.5% a year, reliably positive
Loading on the market	Small, not reliable	Small, not reliable
Loading on company size	Essentially none	Essentially none
Loading on value	Sizeable, negative, reliable	Sizeable, negative, reliable
Share of the month-to-month variation the three forces explain (R-squared)	17.7%	34.4%

What this tells us. In the United States, essentially all of plain momentum's return is already explained by these three forces, mainly a bet against value stocks; the leftover alpha is close to zero and not statistically reliable. In Europe, a large and reliable alpha of about 15.5 percent a year remains after the same three forces are removed, so momentum in Europe was genuinely picking stocks that did well for reasons the three forces cannot explain, not merely riding a wave.

This matters directly for what residualisation is supposed to do. Residualisation strips out exactly the three forces measured here and keeps whatever alpha is left over. In Europe there was substantial alpha to keep, which is why the cleaned up signal there stays healthy. In the United States there was essentially no alpha to keep, only the factor bet residualisation is designed to remove. The cleaning did what it was built to do in both markets; it simply had very different raw material to work with. It is also worth being precise about what this finding is not. The 15.5 percent alpha above is the regression alpha of the plain, uncleaned long-short portfolio after controlling for the three factors; it is not the same object as the 5.2 percent Europe actually earns from residual momentum, which comes from a different procedure entirely, a stock-by-stock rolling regression, a residual mean-to-volatility score, and a fresh cross-sectional ranking each month. Residualising the stocks does not mechanically turn the plain portfolio's alpha into the residual portfolio's return; it only tells us whether there was genuine stock-specific return available for a ranking procedure to find, and in Europe there was, while in the United States there was almost none.

One more question: not what breaks the signal, but when

Every ordinary explanation had failed, and the difference had survived four separate challenges: the sample period, the factor model, the technology sector, and the largest firms. Rather than test a fifth suspect of the same kind, we changed the question. Instead of asking what breaks the United States signal, we asked when it breaks. The shared years split naturally into two very different stretches: 2006 to 2009, when equity markets were in the middle of the financial crisis and its rebound, and 2010 to 2018, the longer, calmer decade that followed. If the United States weakness were only a crisis era phenomenon, the calm decade should look considerably better.

Table 11. Each strategy split into the crisis years and the calm years

Strategy	2006–2009 return (Sharpe)	2010–2018 return (Sharpe)
US Plain	–20.4% (–0.35)	4.6% (0.34)
US Residual	–6.0% (–0.33)	–7.6% (–0.20)
US Volatility targeted	4.5% (0.35)	4.4% (0.43)
EU Plain	–8.5% (–0.02)	23.6% (1.26)
EU Residual	3.6% (0.35)	5.8% (0.81)
EU Volatility targeted	8.5% (0.62)	15.3% (1.17)

What the split shows. Plain momentum crashes and recovers on schedule; residual momentum, in the United States, does not.

Plain momentum, in both markets, behaves exactly as the literature describes: it is hurt badly in the crisis years and recovers cleanly afterwards, United States plain momentum losing 20.4 percent a year in the crisis and gaining 4.6 percent a year once things calmed down, with European plain momentum showing the same shape. This is the well documented pattern of momentum performing badly around market crashes and rebounds (Daniel and Moskowitz, 2016), and it is not unique to either market.

Residual momentum tells a different story, and this is the pivot the rest of this section turns on. In Europe, the cleaned up signal is positive in both stretches, and actually improves in the calm decade, from 3.6 to 5.8 percent a year. In the United States, the cleaned up signal is negative in both stretches, and is, if

anything, slightly worse in the calm decade than in the crisis years, minus 6.0 percent against minus 7.6 percent. So the United States weakness is not a crisis era phenomenon that a calmer decade should have healed. It runs through both halves of the sample. This echoes independent evidence that residual momentum failed to outperform the S&P 500 out of sample over 2011 to 2021 (Pecoraro, 2022).

What was actually happening underneath the calm decade

To understand why a signal with no real underlying alpha (Table 10) still managed to lose 7.6 percent a year in the calm decade, rather than simply sit flat, two further checks were run on that decade alone. The first asked whether the cleaned up score still had any month to month forecasting power at all, measured as the correlation between a stock's score one month and its actual return the next, a measure known in the industry as the information coefficient. The United States and European scores came out with almost identical, weak forecasting power in the calm decade, so the United States score was not uniquely broken; both were behaving like a purified signal with little left to predict. The second asked whether the raw, stock level noise the score is built from was simply larger in the United States, which would mechanically flatten its ranking power. It was not; if anything, European firms showed slightly higher residual volatility than United States firms in the calm decade. Neither of these two more granular checks pointed to anything unusual about the United States on its own. A weak monthly IC is not the same as a weak strategy: a small average ranking edge, applied consistently every month for a decade, can compound into a solid Sharpe ratio even when any single month's correlation is statistically insignificant, which is how Europe's calm-decade residual portfolio earned a 0.81 Sharpe on an IC of only 0.0127. The informative comparison here is not that either market's IC was strong; it is that the United States score matched Europe's rather than falling short of it.

What did stand out was a single month. In August 2011, when the United States government's credit rating was downgraded for the first time in its history and Europe's own sovereign debt crisis flared up again, the United States residual portfolio lost 48 percent, its worst month anywhere in this study. Europe's residual portfolio, in that same month, gained 3.5 percent, so this was not a

shared shock the two markets simply absorbed differently; whatever happened was specific to the United States portfolio. Removing that single month from the calm decade changes the United States residual return from minus 7.6 percent a year to roughly flat, about minus 0.6 percent, while removing the same month from Europe's calm decade return moves it only slightly, from 5.8 to 5.5 percent a year, because no single month in Europe's calm decade plays anything like the same role; its own worst month in the period, January 2012, was a comparatively ordinary minus 5.7 percent, and its best month, June 2016, was plus 7.7 percent. A portfolio built to be market-neutral losing 48 percent in a single month is itself informative: it suggests the rolling 36-month clean cannot fully neutralise a stock's market exposure in exactly the regime-shift months when neutrality matters most, the time-varying exposure problem Grundy and Martin (2001) describe.

What is left, and what stays open

The removals kept failing, and that was itself telling, because it meant the failure was stubborn and real. Every ordinary explanation had been tried, and none of them held, and by itself that would have been a real but incomplete finding, a genuine mystery correctly reported as open. Two further pieces close most of the gap. The raw, uncleaned United States momentum trade carried essentially no genuine stock picking skill in this window, only a bet against value stocks, so residualisation had very little to preserve once it did its job (see Table 10). And in the calmer decade after the financial crisis, when that thin, mostly noise-like signal should simply have sat flat, a single violent month, August 2011, dragged its multi year average down into a clear loss, a month Europe's own signal was not exposed to in the same way.

This is a narrower, more specific finding than “the cause is unknown,” and it is worth being precise about exactly how far it goes. We can say with confidence that United States momentum lacked genuine stock picking skill in this sample, and that a single identifiable month accounts for most of the loss in the years that were supposed to be calm. We cannot say why that skill was absent in this window when earlier decades of momentum research found it, and we cannot say precisely what happened inside the United States portfolio in August 2011 beyond the fact that it coincided with the United States credit downgrade and a

fresh flare up of Europe's sovereign debt crisis, an event Europe's own portfolio was not exposed to in the same way. Those two questions are genuinely open, and they belong in future work, not in a guess here.

The deeper question of why United States momentum lacked skill in this particular window, when it has been found in other decades and other studies, is a larger one than this study can settle, and a discussion should still name the plausible candidates while being clear they are hypotheses, not findings. Market concentration, investor composition, and liquidity and crowding in a heavily arbitrated market remain plausible reasons a momentum trade might lose genuine stock picking skill over time in one market and not another, even though our own tests removing the technology sector and the largest firms did not point to concentration specifically. Each of these would need the wider, multi market study set out in future work to confirm or reject.

What we can say with confidence, then, is this. Over the years we studied, removing the market kept momentum healthy in Europe but broke it in the United States, a gap that survived matching the years, a stronger factor model, removing the technology sector, removing the largest firms, and thousands of reshuffles. Digging further, the United States trade had little genuine skill to begin with, and a single violent month in an otherwise ordinary decade turned a thin, noise-like signal into a clear loss. The broader lesson stands regardless of how the two open questions above are eventually answered: removing the market is not a free improvement everywhere, and treating the cleaned up version as automatically the safer, steadier choice, as the literature tends to, misses markets like the United States where it quietly does harm.

11

Limitations

What this study cannot claim.

This study covers one stretch of time, from 2006 to 2018, because that is the span the data allow. The findings describe that stretch. They cannot, on their own, say what will happen in years outside it.

The concentration question could only be tested by removing groups of companies, first the technology sector and then the largest firms. Both left the result unchanged, which rules those explanations out but does not by itself say what the cause is. With only two markets and data ending in 2018, this study now accounts for much of the failure within the sample: the raw United States signal carried little genuine alpha, and a single month, August 2011, did most of the remaining damage. What stays open is why that alpha was absent in this particular window, and what exactly happened in August 2011; both would need more markets and current data, which we set out in the future work.

The strategies are also frictionless. They ignore the cost of trading, the difficulty and cost of selling shares short, and the trouble of trading the smallest companies. These frictions would lower the returns of all six strategies, and most of all the plain version, which rebuilds its entire top and bottom decile every month and so trades a great deal; a monthly-rebalanced decile long-short strategy of this kind typically turns over most of its positions each month, and even a modest transaction cost of five to ten basis points per trade could plausibly turn plain momentum negative in both markets as well, which would reinforce rather than undermine this study's conclusions. Estimating the exact turnover and cost impact is left for future work. These frictions would not obviously favour one market over the other, but they are real and they are not in the numbers here.

Finally, everything in this study comes from two lists of companies, the S&P 500 and the Stoxx 600, both from a single data provider. Another provider, or a broader list of companies, could make small choices that shift the exact numbers. But the gap between the two markets is large enough that such small

changes would be very unlikely to overturn it. It is simply worth saying plainly that the study looks through one particular window on each market.

12

Future Work

The questions this study opens.

The most useful next step is to bring the data up to date. The sample ends in 2018, and the United States market has changed a great deal since then. It would be valuable to test whether United States residual momentum has recovered in more recent years, especially because one would expect the deepest and most studied market in the world not to stay the weaker of the two for long.

Some of that testing has already begun in this study, which resamples the data thousands of times and runs formal significance tests. The natural next stage is to repeat and extend that work on the updated data: the same resampling, together with Monte Carlo checks, and cross sectional tests that control for company size, so the concentration question can be probed more finely than the sector level test here allowed. Together these would move the finding from a clear pattern in one window to a result that has been stress tested across time and method.

A further important step is to widen the comparison beyond two markets. Running the same test across many markets, such as the United Kingdom, Japan, and the larger emerging markets, would show which behave like Europe and which like the United States, and turn a two market contrast into a pattern that can be studied.

With more markets in hand, the cause can be approached properly. One could line up the markets where residual momentum works against those where it fails, and compare their features to see what actually separates the two groups. That is where a real explanation would come from, rather than from guessing on a single case.

Two narrower questions from this study's own diagnostics deserve a place at the top of that list. The first is why United States momentum's genuine stock picking skill, the return left over once the market, size, and value forces are accounted for, appears to have been essentially absent in this window, when earlier decades of momentum research found it. The second is what specifically happened inside the United States portfolio in August 2011, a month that on its own accounts for most of the signal's loss in an otherwise calm decade, coinciding with the United States credit downgrade and a renewed European sovereign debt scare that Europe's own portfolio was not exposed to in the same way. Both are answerable with more current data and a closer, stock by stock look at that single month, and both would do more to close this study's open question than a further widening of markets alone.

None of this offers a fix, and this study does not pretend to have one. What it offers is a clear map of how the cause could be found, which is the honest next stage for a question this study was able to open but not to close.

13

Conclusion

When a cleaner signal is not the better one.

We set out expecting the cleaned up version of momentum to be the better one, as the literature suggests. Europe agreed. The United States did not. The same cleaned up signal that behaves well in Europe was weak in the United States, and the gap was large enough to demand an explanation.

We first suspected the United States simply had more years of data, so we cut both markets to the same years. The gap did not shrink. It grew, and the United States signal turned negative. So the time period was not the reason.

We then removed the market entirely and looked at what was left. In the United States the leftover lost money and behaved close to random, with a worst month of nearly minus 48 percent. In Europe the leftover stayed healthy. So the difference is structural, and not a product of the years we studied.

We then asked whether we had removed the market too simply, and removed five forces instead of three. The United States improved but still did not make money, while Europe still won. So the difference is not the method either.

We then pressed the result harder still. We reshuffled the months into thousands of alternative histories, and Europe beat the United States in almost every one, so the gap is not the luck of a single sample. We removed the entire technology sector, and then the largest firms of all, in case a few giant companies were the cause, and the United States signal still did not recover. Every ordinary explanation had been ruled out, and the difference held.

Ruling out every ordinary explanation was not the end of the investigation. We went back to the raw, uncleaned United States signal and asked whether it had ever carried genuine stock picking skill; it had not, its return was almost entirely a bet against value stocks. We then split the shared years into the crisis period and the calm decade that followed, and found the United States weakness ran through both, not just the crisis. Digging into the calm decade specifically, a single month, August 2011, accounted for most of its loss, a month Europe's own signal passed through largely unaffected.

The conclusion is careful and specific. Over 2006 to 2018, with this dataset and these methods, residual momentum is positive in Europe but not in the United States, and that gap survives matching the years, a stronger factor model, removing the technology sector, removing the largest firms, and thousands of reshuffles. So residual momentum is not the same in every market: the cleaned up version the literature treats as the safer, steadier choice can, as in the United States over this period, quietly fail to deliver. Within this study the difference is real and structural, and we can now say more than that it is structural: the United States trade lacked genuine skill to begin with, and one violent month turned a thin, noise-like signal into a clear loss. What remains genuinely open is why that skill was absent in this window, and what exactly happened in August 2011, and settling those two questions, alongside widening the study to more markets and to current data, is where the next work begins.

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15

Appendix: Data and Methodology

The data, every processing decision, and the two supplementary diagnostics, documented for replication.

This appendix documents the data behind the study and every decision taken in building it, at a level of detail that would let someone starting from the same raw files reproduce the same numbers. It uses standard financial-econometrics terminology throughout, since that is the register in which this kind of work is normally documented and reviewed; a short plain-language gloss is given the first time each technical term appears.

A. Data inventory

Five datasets went into the study, two supplied by the course and three drawn from the public Kenneth French data library.

Table A1. Data sources

Dataset	Coverage and content	Source	Notes
United States panel	S&P 500 constituents, monthly prices, 1990 to 2018, 1,355 firms, survivorship free	Course dataset (e-learning platform), Variant C brief	Includes firms that later left, merged, or failed. Not independently verifiable
European panel	Stoxx 600 constituents, monthly prices, 2002 to 2018, 1,255 firms, survivorship free	Course dataset (e-learning platform), Variant C brief	Same caveat. Sector sheets used only for the technology exclusion test
Fama-French factors, US	Market, Size, Value (MKT-RF, SMB, HML), plus Profitability and Investment (RMW, CMA) for the five-factor test; monthly, net of the risk-free rate	Kenneth French library (public)	US dollar basis
Fama-French factors, Europe	The same factors on a European basis, monthly, net of the European risk-free rate	Kenneth French library (public)	Euro basis, matching the European panel
Risk-free rates	Monthly risk-free series embedded in the Kenneth French factor files	Kenneth French library (public)	US series with the US panel, European series with the European panel

Currency basis. United States prices, returns, and factors are in US dollars throughout; European prices, returns, and factors are in euros throughout. No

cross-currency conversion was applied at any point, and none was needed: every comparison in the study is between a market's own strategies over its own history, or between the two markets' summary statistics (returns, Sharpe ratios, and the like), never a direct dollar-for-dollar or euro-for-dollar comparison of raw cash values.

B. Processing decisions

Every discretionary choice made in building the six strategies is listed below, in the order the main text uses them.

Universe construction. The study keeps every firm that was ever a member of the S&P 500 or the Stoxx 600, not only the firms in each index today (Section 3). Membership windows come from the course dataset.

Returns. Monthly returns were computed as simple discrete returns (`R` `Return.calculate, method = "discrete"`) directly from the "Price" sheet of each course-supplied workbook. No separate dividend or corporate-action adjustment was applied beyond whatever the provider's price series already reflects, and no additional cleaning, such as forward-filling gaps or merging duplicate share classes, was performed. A missing price simply produces a missing return for that firm-month, which is handled by the missing-months rule below.

Missing months. A firm with no return in a given month is left out of that month's ranking rather than filled in with an imputed value (Section 4).

Plain momentum score. The twelve-month compounded return, skipping the most recent month: $M(i,t) = (1 + r(t-12)) \times \dots \times (1 + r(t-2)) - 1$ (Section 4).

Residualisation. For each firm, an OLS regression of returns on the three, or five, factors over a rolling thirty-six month window; the residual momentum score is the mean of the last twelve residuals from that regression (the formation window, skipping the most recent month), divided by their standard deviation over the same window. This ratio is a monthly information ratio: it rewards a stock whose company-specific return, net of the market, size, and value forces, has been both positive and consistent. A firm with fewer than thirty-six valid months in the estimation window receives no score that month,

which is why the European residual sample effectively begins in 2006 (Section 4).

Pseudocode: residual momentum score construction (Section 4)

```
for each month t:
  for each firm i:
    if firm i has fewer than 36 valid months of returns:
      score is missing; skip firm this month
    else:
      regress firm i's excess returns on [Market, Size, Value]
      over the last 36 months
      keep the residuals e
      score(i,t) = mean(e over last 12 months, skipping the
                        most recent) / std(same 12 residuals)
rank all scored firms; top decile = long leg (equal weight),
bottom decile = short leg (equal weight); hold one month
```

Factor models. The Fama-French three-factor model (market, size, value) for the baseline; the five-factor model (adding profitability and investment) for the robustness test in Section 9.

Portfolio construction. Each month, firms are ranked into deciles by the previous month's score; the top decile is held long and the bottom short, equal-weighted within each leg. Portfolios are rebuilt monthly and are non-overlapping (Section 4).

Volatility-targeted variant. The plain momentum portfolio, scaled by $w(t) = \text{target risk} / \text{trailing realised volatility of the long-short portfolio over the previous six months}$ (Section 4).

Performance metrics. Annualised return, annualised volatility (the monthly standard deviation scaled by the square root of twelve), the Sharpe ratio (annualised return over annualised volatility, using a zero risk-free rate for the long-short portfolios, which are self-financing), and maximum drawdown from a running peak (Section 4).

Matched window. Every cross-market comparison runs on 2006 to 2018, the years both markets share once residual momentum's thirty-six month burn-in is satisfied (Section 7).

Statistical tests. A binomial test of the share of positive months against one half, and a one-sample t-test of the mean monthly return against zero, both on the matched window (Section 8). An optional Newey-West (HAC) robustness

check was also run on the mean-return test, using a fixed six-month lag in one version and the sandwich package's automatic lag selection in another; neither changes the conclusion, so the primary text reports the unadjusted t-test.

Bootstrap. A moving-block bootstrap with six-month blocks and 5,000 replications, drawn jointly from the same block starting points for both markets so that their month-to-month co-movement is preserved and the gap between them can itself be bounded, on the matched window (Section 8).

Pseudocode: moving-block bootstrap (Section 8)

```
repeat 5,000 times:
  draw a resample of six-month blocks from the shared years,
    choosing the same block starting points for both markets
  compute each market's Sharpe ratio on the resample
report: each market's median Sharpe, its 95 percent range,
  and the share of resamples in which Europe beat the US
```

Sector and size exclusions. The technology-exclusion test drops every GICS or ICB technology firm, 197 in the United States and 57 in Europe; the size-exclusion test drops each month's largest firms by index weight (Section 10).

Two supplementary diagnostics were added after the first draft, prompted by the observation that ruling out every ordinary explanation identifies what the United States shortfall is not, without identifying what it is. Both use only the data already assembled above.

Factor decomposition of plain momentum. Each market's uncleaned (plain) monthly long-short momentum return was regressed on its own three Fama-French factors over the matched window: $R(t) = \alpha + \beta_1 \cdot \text{MktRF}(t) + \beta_2 \cdot \text{SMB}(t) + \beta_3 \cdot \text{HML}(t) + \varepsilon(t)$, with Newey-West standard errors (lag 6). Alpha (α) is the average monthly return the three factors cannot explain — the closest available proxy for genuine stock-selection skill, as opposed to a bet on a known, rewarded risk factor. Results, matched window: United States — $\alpha = 0.00003$ ($t = 0.004$, not significant), $\beta(\text{MktRF}) = -0.0020$ ($t = -0.87$), $\beta(\text{SMB}) = 0.0001$ ($t = 0.04$), $\beta(\text{HML}) = -0.0109$ ($t = -3.08$, $p = 0.002$), $R^2 = 0.177$, annualised alpha $\approx 0.0\%$. Europe — $\alpha = 0.0129$ ($t = 2.26$, $p = 0.025$), $\beta(\text{MktRF}) = -0.0024$ ($t = -1.46$), $\beta(\text{SMB}) = -0.0016$ ($t = -0.67$), $\beta(\text{HML}) = -0.0148$ ($t = -5.19$, $p < 0.001$), $R^2 = 0.344$, annualised alpha $\approx 15.5\%$. Reading: United States plain momentum carried essentially no alpha and was almost entirely a (loss-making) bet against

value stocks; European plain momentum carried large, statistically reliable alpha in addition to a similar value bet. Since residualisation is designed to remove factor exposure and retain alpha, this explains why the cleaned signal remained healthy in Europe (there was alpha to retain) and collapsed toward noise in the United States (there was almost none).

Pseudocode: factor decomposition, the alpha test (Table 10)

```
for each market:
    collect the plain long-short portfolio's monthly returns
        over the shared years
    regress these returns on [MktRF, SMB, HML]
        with Newey-West standard errors
    report: alpha (x 12 for annualised), each beta with its
        t-statistic, and R-squared
```

Sub-period split. The matched window was split at the end of 2009 into a crisis sub-period (February 2006 to December 2009, $n = 47$ months) and a calm sub-period (January 2010 to November 2018, $n = 107$ months), and all six strategies were re-run on each slice using the same performance metrics as Section 4.

Table A2. Sub-period statistics

Strategy	Period	AnnRet	AnnVol	Sharpe	MaxDD
US Plain	2006–2009	−20.4%	38.2%	−0.35	78.1%
US Plain	2010–2018	4.6%	17.4%	0.34	36.0%
US Residual	2006–2009	−6.0%	15.4%	−0.33	37.7%
US Residual	2010–2018	−7.6%	22.3%	−0.20	69.9%
EU Plain	2006–2009	−8.5%	35.5%	−0.02	74.1%
EU Plain	2010–2018	23.6%	18.3%	1.26	20.7%
EU Residual	2006–2009	3.6%	12.4%	0.35	26.2%
EU Residual	2010–2018	5.8%	7.3%	0.81	14.7%

Reading. United States plain momentum shows the textbook momentum-crash pattern (severe loss in the crisis sub-period, recovery in the calm sub-period). United States residual momentum does not follow that pattern: it is negative in both sub-periods and marginally worse in the calm sub-period, indicating the shortfall is not confined to the 2008 financial crisis window.

Information coefficient (IC). The IC is the Spearman rank correlation, computed cross-sectionally each month, between a stock's residual momentum score (formed at $t-1$) and its realised return at t ; it measures the score's

forward-looking rank predictive power. Averaged over the calm sub-period, mean IC was 0.0132 ($t = 1.52$, $n = 106$) for the United States and 0.0127 ($t = 1.55$, $n = 106$) for Europe — statistically indistinguishable between markets, and neither individually significant at conventional levels. This rules out the specific hypothesis that the United States score had uniquely deteriorated month-to-month predictive power in the calm sub-period.

Firm-level residual volatility. The standard deviation of the same rolling-regression residuals used to build the momentum score, computed at the individual-firm level rather than aggregated into the score. Averaged over the calm sub-period, mean monthly firm-level residual volatility was 4.27% for the United States and 5.73% for Europe — higher, not lower, in Europe. This rules out the hypothesis that a mechanically larger noise term was flattening the United States score's ranking power relative to Europe's.

The August 2011 month. United States residual momentum's single worst month in the full sample was August 2011 (−48.0%), coinciding with the first-ever downgrade of the United States sovereign credit rating and a renewed flare-up of the European sovereign-debt crisis. European residual momentum's return in the same month was +3.5%, and its own worst month in the calm sub-period, January 2012 (−5.7%), and best month, June 2016 (+7.7%), are both unremarkable by comparison. Removing August 2011 from the United States calm-sub-period series changes its annualised return from −7.6% to −0.6% (Sharpe from −0.20 to 0.06, maximum drawdown from 69.9% to 42.1%); removing the same month from Europe's calm-sub-period series changes its annualised return only marginally, from 5.8% to 5.5%, with maximum drawdown unchanged. This indicates a single, market-specific event, rather than a shared shock, accounts for most of the United States calm-sub-period loss.

C. Source documents

Variant C course brief, Financial Markets Analytics 2026 (Forte), sections 4 to 5, which specifies the two course datasets and points to the Kenneth French data library for factors.

Main project document, Residual Momentum Across Markets, sections 3, 4, 7, 8, 9, 10 and 11.